

FLOATING POINT REPRESENTATION

PRACTICE SHEET • MEDIUM LEVEL / SET 2

ResourceOS

1. What is the 32-bit single-precision IEEE 754 hexadecimal format for decimal fraction +0.15625?

- (A) 3E200000 (B) 3E500000
(C) 3F200000 (D) BE200000

2. In an 8-bit floating-point format (1 sign bit, 3 exponent bits with bias = 3, and 4 mantissa bits), what is the decimal value of 0 101 0100?

- (A) 5.0 (B) 2.5
(C) 10.0 (D) 1.25

3. What is the unbiased exponent for the largest positive normalized value in IEEE 754 single precision?

- (A) +127 (B) +128
(C) +254 (D) +126

4. In IEEE 754 half-precision (16-bit: 1 sign, 5 exponent with bias = 15, 10 mantissa), what stored exponent represents subnormal numbers?

- (A) 00000_2 (B) 00001_2
(C) 11111_2 (D) 01111_2

5. What is the machine epsilon (ϵ) value for IEEE 754 double precision (64-bit)?

- (A) 2^{-52} (B) 2^{-53}
(C) 2^{-23} (D) 2^{-1023}

6. What is the decimal value of the IEEE 754 single-precision representation 0xC0400000?

- (A) -3.0 (B) -1.5
(C) +3.0 (D) -6.0

7. During floating-point addition, what step must occur before adding the two mantissas?

- (A) Alignment of exponents (shifting smaller mantissa right) (B) Multiplication of exponents
(C) Rounding the larger mantissa (D) Negating the sign bits

8. Which bit pattern distinguishes a Quiet NaN (QNaN) from a Signaling NaN (SNaN) in IEEE 754 standard?

- (A) MSB of the fraction field is 1 for QNaN and 0 for SNaN (B) MSB of the fraction field is 0 for QNaN and 1 for SNaN
(C) Sign bit is 1 for QNaN and 0 for SNaN (D) Exponent is all 1s for QNaN and all 0s for SNaN

9. Why cannot decimal 0.2 be represented with exact precision in binary floating-point systems?

- (A) Its binary representation is an infinitely repeating fraction (0.00110011...) (B) Its exponent exceeds the single precision exponent bias
(C) It is smaller than the smallest subnormal floating point number (D) It requires a negative bias factor

10. Using IEEE 754 default rounding (Round to Nearest, Ties to Even), round the binary value 1.010100_2 to 3 fractional bits.

- (A) 1.010_2 (B) 1.011_2
(C) 1.100_2 (D) 1.001_2

11. In IEEE 754 single precision, what is the value of the 23-bit mantissa field for decimal 1.5?

- (A) $10000000000000000000000_2$ (B) $01000000000000000000000_2$
(C) $11000000000000000000000_2$ (D) $00000000000000000000000_2$

12. What is the exponent bias for a custom 10-bit floating-point format that allocates 4 bits for exponent?

- (A) 7 (B) 15
(C) 8 (D) 31

13. Express decimal fraction -0.03125 in IEEE 754 single-precision normalized scientific notation.

- (A) $-1.0_2 \times 2^{-5}$ (B) $-1.0_2 \times 2^{-4}$
(C) $-0.1_2 \times 2^{-5}$ (D) $-1.0_2 \times 2^{-6}$

14. In single precision, what stored 8-bit binary exponent corresponds to unbiased exponent $E = -5$?

- (A) 01111010_2 (122) (B) 01111011_2 (123)
(C) 10000100_2 (132) (D) 01111001_2 (121)

15. What numerical evaluation error occurs when subtracting two floating-point numbers that are extremely close in magnitude?

- (A) Loss of significance / Catastrophic cancellation (B) Exponent overflow
(C) Gradual underflow error (D) Division-by-zero trap

16. Which formula correctly computes the value of a subnormal number in single-precision IEEE 754?

- (A) $(-1)^S \times (0.M) \times 2^{-126}$ (B) $(-1)^S \times (1.M) \times 2^{-127}$
(C) $(-1)^S \times (0.M) \times 2^{-127}$ (D) $(-1)^S \times (1.M) \times 2^{-126}$

17. Convert decimal integer 255 to single-precision IEEE 754 binary normalized mantissa.

- (A) $11111110000000000000000_2$ (B) $11111111000000000000000_2$
(C) $01111111000000000000000_2$ (D) $11111101000000000000000_2$

18. What is the stored 8-bit binary exponent for decimal integer 255 in IEEE 754 single precision?

- (A) 10000110_2 (134) (B) 10000111_2 (135)
(C) 01111111_2 (127) (D) 10000101_2 (133)

19. What is the IEEE 754 standard result of evaluating expression $\text{sqrt}(-1.0)$?

- (A) NaN (B) -1.0
(C) -Infinity (D) Subnormal zero

20. What is the effect of applying the IEEE 754 "Round toward Zero" (truncation) mode on negative numbers?

- (A) Rounds toward greater (less negative) values (B) Rounds toward smaller (more negative) values
(C) Always rounds to nearest even integer (D) Behaves identically to round toward negative infinity

21. How many total bits are used in IEEE 754 Quadruple Precision format?

- (A) 128 bits (B) 80 bits
(C) 64 bits (D) 256 bits

22. In single precision, what does the hex bit sequence 0x00400000 represent?

- (A) Smallest positive normalized number (B) A subnormal (denormalized) number
(C) Zero (D) Signaling NaN

23. What is the outcome of floating-point comparison $\text{NaN} == \text{NaN}$ in IEEE 754 compliant systems?

- (A) False (B) True
(C) Exception Trap (D) Undefined behavior

24. Convert decimal 0.05 to binary fraction. What is its repeating pattern?

- (A) 0.000011001100...₂ (1100 repeats) (B) 0.00110011...₂ (0011 repeats)
(C) 0.01010101...₂ (01 repeats) (D) 0.00010101...₂ (01 repeats)

25. In IEEE 754 double precision (64-bit), how many bits are allocated for the exponent field?

- (A) 11 bits (B) 8 bits
(C) 15 bits (D) 52 bits

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	6	(A)	11	(A)	16	(A)	21	(A)
2	(A)	7	(A)	12	(A)	17	(A)	22	(A)
3	(A)	8	(A)	13	(A)	18	(A)	23	(A)
4	(A)	9	(A)	14	(A)	19	(A)	24	(A)
5	(A)	10	(A)	15	(A)	20	(A)	25	(A)